

# CSE 350, Summer 2024

Dear all, Please submit your questions in this document. If you want to submit them in Latex, just drop the link for the Latex file here and attach a screenshot of the question.

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DTL: Himaddri Roy , Jishnu Mahmud

TTL: Quazi Shafayat Hossain

Please provide a step-by-step solution at the same time as the question—deadline **6th June 2024**.

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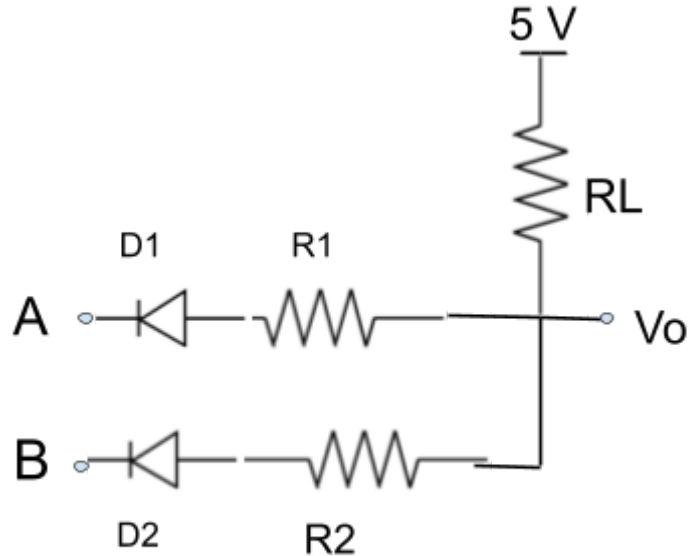
- While posting your question here, please follow this format:

|     |    |          |         |
|-----|----|----------|---------|
| (a) | CO | Question | [Marks] |
| (b) | CO | Question | [Marks] |
| (c) | CO | Question | [Marks] |

DL:[TNMF]

R this calculation)SET A

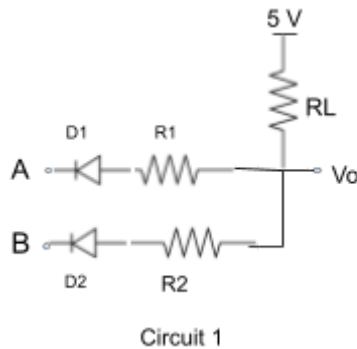
|       |     |                                                                                                                                                                                                                                                           |       |            |            |       |
|-------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------|------------|-------|
| 1.CO1 | (a) | For the given DL circuit-1, assume $V_{DD}$ (logic 1)=5V, $V$ (Logic 0)=0.2V, $R1=R2= 200\ \Omega$ and $RL= 150\ k\Omega$ . Find the Output voltage, $V_o$ for all the input combinations and fill out the given table. Identify the gate from the table. |       |            |            | [8+2] |
|       |     | $V_A$                                                                                                                                                                                                                                                     | $V_B$ | D1[On/OFF] | D2[On/OFF] | $V_o$ |
|       |     |                                                                                                                                                                                                                                                           |       |            |            |       |
|       |     |                                                                                                                                                                                                                                                           |       |            |            |       |
|       |     |                                                                                                                                                                                                                                                           |       |            |            |       |
|       | (b) | (Assume, the output voltage $V_o$ from the table is fixed for $V_A=V_B=0.2\ V$ ). If the value of <i>maximum</i> current is <i>doubled</i> , find the change in the resistor value. Comment on which resistor value will be changed.                      |       |            |            | [4+1] |
|       | (c) | Calculate the minimum value of R in $K\Omega$ , if we want to keep the maximum power dissipation under 0.5mW. [hint: You may neglect the value of $R1$ or $R2$ as they are very small compared to R.                                                      |       |            |            | [5]   |



Circuit 1

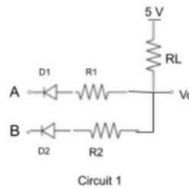
## SET B

|       |     |                                                                                                                                                                                                                                                                                  |       |            |            |       |
|-------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------|------------|-------|
| 1.CO1 | (a) | For the given DL circuit-1, assume $V_{DD}(\text{logic } 1)=5V$ , $V(\text{Logic } 0)=0.2V$ , $R1=R2= 300\ \Omega$ and $RL= 100\ k\Omega$ . Find the Output voltage, $V_o$ for all the input combinations and fill out the given table. <b>Identify</b> the gate from the table. |       |            |            | [8+2] |
|       |     | $V_A$                                                                                                                                                                                                                                                                            | $V_B$ | D1[On/OFF] | D2[On/OFF] | $V_o$ |
|       |     |                                                                                                                                                                                                                                                                                  |       |            |            |       |
|       |     |                                                                                                                                                                                                                                                                                  |       |            |            |       |
|       |     |                                                                                                                                                                                                                                                                                  |       |            |            |       |
|       | (b) | Assume, the output voltage $V_o$ from the table is fixed for $V_A=V_B=0.2\ V$ . If the value of <i>maximum</i> current is <i>doubled</i> , <b>find</b> the change in the resistor value. Comment on which resistor value will be changed.                                        |       |            |            | [4+1] |
|       | (c) | <b>Calculate</b> the minimum value of $R$ in $K\Omega$ , if we want to keep the maximum power dissipation under $0.5mW$ . [hint: You may neglect the value of $R1$ or $R2$ as they are very small compared to $R$ .]                                                             |       |            |            | [5]   |



Solution:

|       |     |                                                                                                                                                                                                                                                                 |       |            |            |       |       |
|-------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------|------------|-------|-------|
| 1.CO1 | (a) | For the given DL circuit-1, assume $V_{DD}$ (logic 1)=5V, $V(\text{Logic } 0)$ =0.2V, $R1=R2=200\ \Omega$ and $RL=150\ k\Omega$ . Find the Output voltage, $V_o$ for all the input combinations and fill out the given table. Identify the gate from the table. |       |            |            | [8+2] |       |
|       |     | $V_A$                                                                                                                                                                                                                                                           | $V_B$ | D1[On/OFF] | D2[On/OFF] |       | $V_o$ |
|       |     |                                                                                                                                                                                                                                                                 |       |            |            |       |       |
|       |     |                                                                                                                                                                                                                                                                 |       |            |            |       |       |
|       |     |                                                                                                                                                                                                                                                                 |       |            |            |       |       |
|       |     |                                                                                                                                                                                                                                                                 |       |            |            |       |       |
|       |     |                                                                                                                                                                                                                                                                 |       |            |            |       |       |
|       | (b) | Assume, the output voltage $V_o$ from the table is fixed for $V_A=V_B=0.2\text{ V}$ . If the value of maximum current is doubled, find the change in the resistor value. Comment on which resistor value will be changed.                                       |       |            |            | [4+1] |       |
|       | (c) | Calculate the minimum value of R in $K\Omega$ , if we want to keep the maximum power dissipation under 0.5mW. [hint: You may neglect the value of $R_1$ or $R_2$ as they are very small compared to R.]                                                         |       |            |            | [5]   |       |



SET A

$$R_1=R_2=200\ \Omega; R_L=150\ k\Omega$$

a) [8 marks]

Case I

$$V_A=V_B=0.2\ V$$

$$2I_1=I$$

$$2 \times \frac{V_o - 0.9}{0.2\ k\Omega} = \frac{5 - V_o}{150\ k\Omega}$$

$$\Rightarrow V_o = 0.903\ V$$

Case II & III

$$V_{A,B}=0.2; V_{R,L}=5\ V$$

$$I_1=I$$

$$\frac{V_o - 0.9}{0.2\ k\Omega} = \frac{5 - V_o}{150\ k\Omega}$$

$$\Rightarrow V_o = 0.91\ V$$

Case W

$$V_A=V_B=5\ V$$

$$I=0; V_o=5\ V$$

SET B

$$R_1=R_2=300\ \Omega; R_L=100\ k\Omega$$

a) [8 marks]

Case I

$$V_A=V_B=0.2\ V$$

$$2I_1=I$$

$$2 \times \frac{V_o - 0.9}{0.3\ k\Omega} = \frac{5 - V_o}{100\ k\Omega}$$

$$\Rightarrow V_o = 0.906\ V$$

Case II & III

$$V_{A,B}=0.2; V_{R,L}=5\ V$$

$$I_1=I$$

$$\frac{V_o - 0.9}{0.3\ k\Omega} = \frac{5 - V_o}{100\ k\Omega}$$

$$\Rightarrow V_o = 0.912\ V$$

Case W

$$V_A=V_B=5\ V$$

$$I=0; V_o=5\ V$$

Judging from the table, it's a DL AND gate. [1 marks]

b)  $V_o = 0.903V$  for  $V_A = V_B = 0.2V$

$$V_R = 5 - 0.903V = 4.097V$$

As,  $V_{R_{new}} = V_{R_{old}}$

$$\Rightarrow I_{max} \cdot R_{L0} = 2 \times I_{max} \cdot R_{L_{new}}$$

$$\Rightarrow R_{L_{new}} = \frac{I_{max}}{2 \times I_{max}} \times R_{L_{old}}$$

$$\Rightarrow R_{L_{new}} = \frac{1}{2} \times R_{L_{old}}$$

$$R_{L_{new}} = 75k\Omega \quad [4 \text{ marks}]$$

$$R_{1_{new}} = R_{2_{new}} = 100\Omega \quad [4 \text{ marks}]$$

c)  $P = VI \Rightarrow (5 - 0.2) \frac{5 - V_o}{R_L} < 0.5mW$

$$\Rightarrow R_L \geq 39.33k\Omega \quad [5 \text{ marks}] \quad [R_1, R_2 \text{ are}$$

neglected to simplify the calculation]  
If  $R_1$  &  $R_2$  is considered, 'P' can be deduced as a function of  $R_L, R_1$  &  $R_2$  and same  $R_L$  will be found by comparing them. However, it will complicate the calculation as  $V_o$  is also unknown.]

Judging from the table, it's a DL AND gate. [1 marks]

b)  $V_o = 0.90V$  for  $V_A = V_B = 0.2V$

$$V_R = 5 - 0.906V = 4.094V$$

As,  $V_{R_{new}} = V_{R_{old}}$

$$\Rightarrow I_{max} \cdot R_{L0} = 2 \times I_{max} \cdot R_{L_{new}}$$

$$\Rightarrow R_{L_{new}} = \frac{I_{max}}{2 \times I_{max}} \times R_{L_{old}}$$

$$\Rightarrow R_{L_{new}} = \frac{1}{2} \times R_{L_{old}}$$

$$R_{L_{new}} = 50k\Omega \quad [4 \text{ marks}]$$

$$R_{1_{new}} = R_{2_{new}} = 150\Omega \quad [4 \text{ marks}]$$

c)  $P = VI \Rightarrow (5 - 0.2) \frac{5 - V_o}{R_L} < 0.5mW$

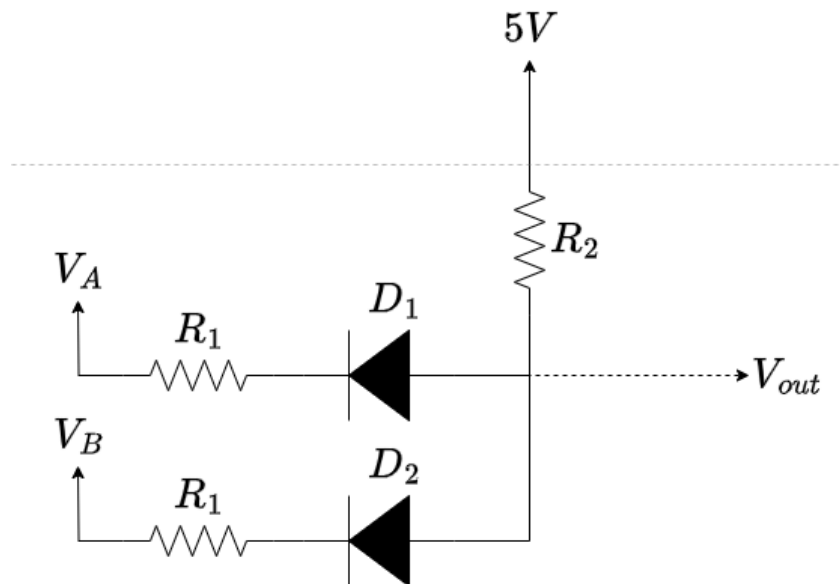
$$\Rightarrow R_L \geq 39.30k\Omega \quad [5 \text{ marks}]$$

$$R_{min} = 39.30k\Omega$$

**DL: [TMY]**

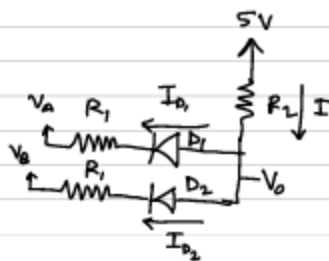
Set - A

For the logic circuit in figure 1, assume input voltages of **VA**, **VB** can be **0V / 5V** . Also assume that **1 kΩ** and **60 kΩ** value resistors are available to be placed in the following circuit.



**Figure: Circuit - 1**

- 1) **(CO2)** From the resistor values available, determine the appropriate resistor values to be used for  $R_1$  and  $R_2$  so that the following circuit acts as a logic **AND** gate (*N.B: None of the resistor values can be set to 0*). Justify your design by finding the output voltages of the circuit for all possible input voltage combinations. **[10]**
- 2) **(CO2)** For your designed circuit, find out the maximum power dissipation. **[10]**



Set-A:  $R_1 = 1k, R_2 = 60k$

①

Case 1:  $V_A = 0V, V_B = 0V$ :

$D_1, D_2$  ON: Node at  $V_0$ :

$$\frac{V_0 - 5}{60} + \frac{(V_0 - 0.7) - 0}{1} \times 2 = 0$$

$$\Rightarrow V_0 - 5 + 120V_0 - 84 = 0$$

$$\Rightarrow V_0 = 0.7355V$$

$$I = \frac{5 - 0.7355}{60} = 0.071mA$$

$$I_{D1} = I_{D2} = \frac{I}{2} = 0.0355mA$$

Case 2+3:

$V_A = 5V, V_B = 0V$

or,  
 $(V_A = 0V, V_B = 5V)$

$D_1$  OFF,  $D_2$  ON:

$$\frac{V_0 - 5}{60} + \frac{(V_0 - 0.7) - 0}{1} = 0$$

$$\dots \Rightarrow V_0 = \frac{17}{60} = 0.7833V$$

$$I_{D1} = 0, I_{D2} = \frac{(0.7833 - 0.7)}{1k}$$

$$= 0.0833mA$$

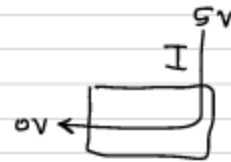
$$I = I_{D2}$$

Case 4:  $V_A = 5V, V_B = 5V \rightarrow D_1, D_2$  OFF  $\rightarrow I_{D1} = I_{D2} = 0mA$

$\therefore V_0 = 5V$

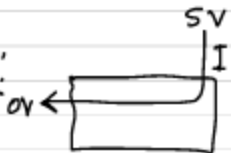
②

Case ①:



$$I = 0.071mA, P = (5 - 0) \times I = 0.355mW$$

Case ②+③:



$$I = 0.083mA, P = (5 - 0) \times I = 0.415mW$$

Case ④:



No current.  $P = 0$

## Set - B

For the logic circuit in figure 1, assume input voltages of  $V_A$ ,  $V_B$  can be  $0V/5V$ . Also assume that  $570\ \Omega$  and  $80\ k\Omega$  value resistors are available to be placed in the following circuit.

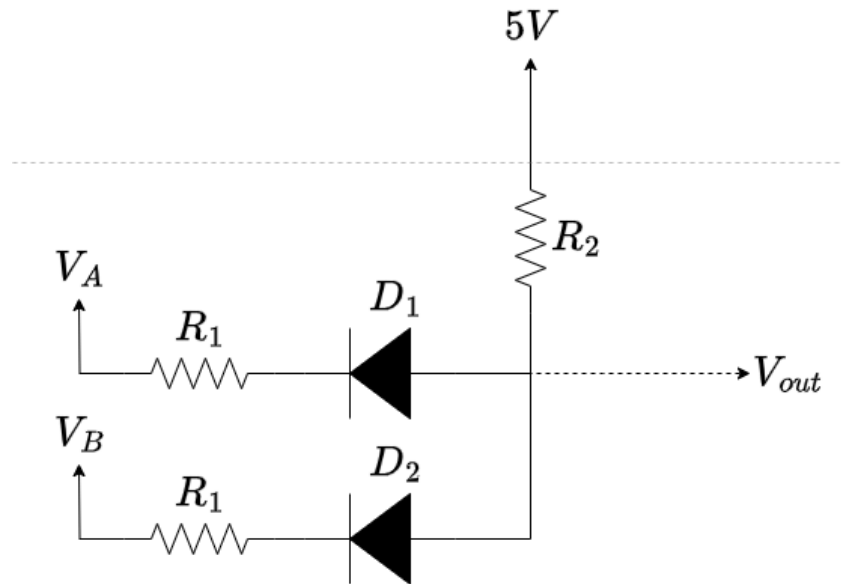
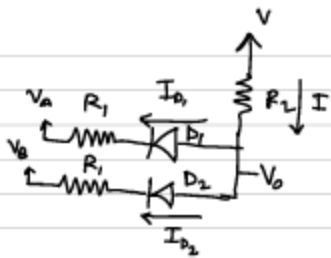


Figure: Circuit - 1

- 3) **(CO2)** From the resistor values available, determine the appropriate resistor values to be used for  $R_1$  and  $R_2$  so that the following circuit acts as a logic **AND** gate (*N.B: None of the resistor values can be set to 0*). Justify your design by finding the output voltages of the circuit for all possible input voltage combinations. **[10]**
- 4) **(CO2)** For your designed circuit, find out the maximum power dissipation. **[10]**



Set-B:  $R_1 = 570, R_2 = 80k$   
 $= 0.57k$

①

Case 1:  $V_A = 0V, V_B = 0V$ :

$D_1, D_2$  ON: Node at  $V_0$ :

$$\frac{V_0 - 5}{80} + \frac{(V_0 - 0.7) - 0}{0.57} \times 2 = 0$$

$$\Rightarrow V_0 \left( \frac{1}{80} + \frac{2}{0.57} \right) = \frac{5}{80} + \frac{0.7 \times 2}{0.57}$$

$$\Rightarrow V_0 = 0.715V$$

$$I = \frac{5 - 0.715}{80} = 0.0536mA$$

$$I_{D1} = I_{D2} = \frac{I}{2} = 0.0268mA$$

Case 2+3:

$$V_A = 5V, V_B = 0V$$

$$\left( \begin{array}{l} \text{or} \\ V_A = 0V, V_B = 5V \end{array} \right)$$

$D_1$  OFF,  $D_2$  ON:

$$\frac{V_0 - 5}{80} + \frac{(V_0 - 0.7) - 0}{0.57} = 0$$

$$\dots \Rightarrow V_0 = \frac{\frac{5}{80} + \frac{0.7}{0.57}}{\left( \frac{1}{80} + \frac{1}{0.57} \right)}$$

$$\therefore V_0 = 0.73V$$

$$I_{D1} = 0, I_{D2} = \frac{(0.73 - 0.7)}{0.57}$$

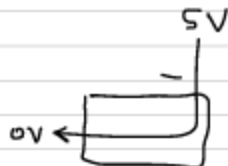
$$I = I_{D2} = 0.053mA$$

Case 4:  $V_A = 5V, V_B = 5V \rightarrow D_1, D_2$  OFF  $\rightarrow I_{D1} = I_{D2} = 0mA$

$$\therefore V_0 = 5V$$

②

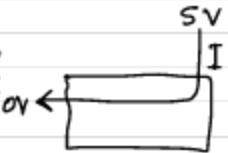
Case ①:



$$I = 0.0536mA, P = (5 - 0) \times I$$

$$= 0.268mW$$

Case ②+③:



$$I = 0.053mA, P = (5 - 0) \times I$$

$$= 0.265mW$$

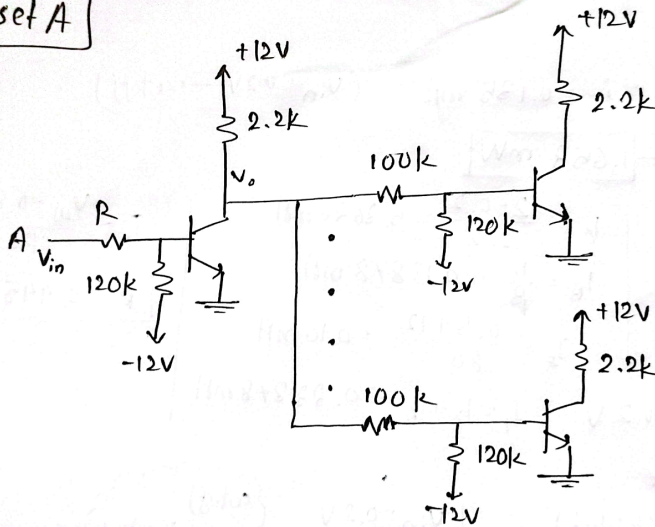
Case ④:



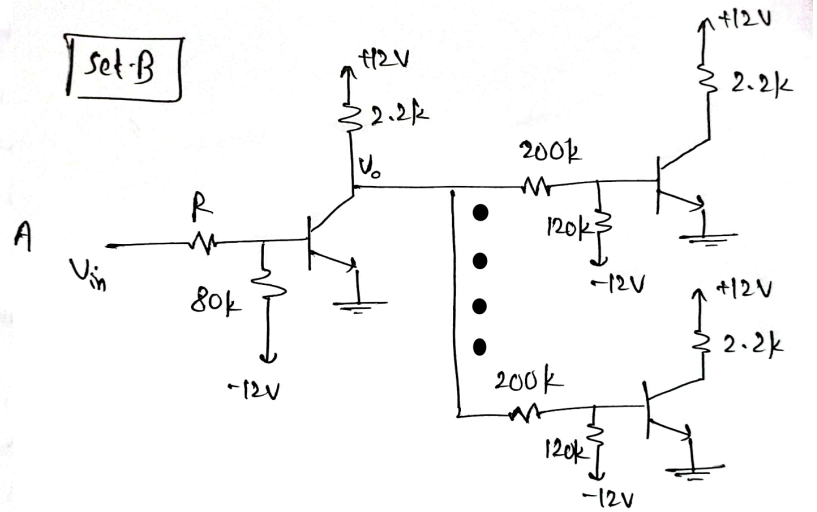
No current.  $P = 0$

# RTL:[RDM]

set A



set B



Assume,

$$V_{BE(sat)} = 0.8V, V_{CE(sat)} = 0.2V, \beta_F = 30, V_T = 0.5V, V_{OH} = 10V, V_{OL} = 0.2V.$$

(a) - c02 - (a) Consider no loads are attached to driver circuit, find total power dissipated in the driver when input,  $V_{in} = 0.2V$  (assume  $R = 10k\Omega$ ) [4]

c02 - (b) For high noise margin ( $V_{NH}$ ) of 6V, estimate the value of  $R$  in  $k\Omega$  when no loads are connected to the driver. [8]

c01 - (c) Determine the maximum possible fanout (assume  $R = 10k\Omega$ ) [8]

Solution:

### Solution Set A

$$(a) \frac{0.2 - (-12)}{10 + 120} = 0.0938 \text{ mA} = i \quad (V_{in} = 0.2 \text{ V} \rightarrow \text{cutoff})$$

$$P = (0.2 + 12) i = 1.1449 \text{ mW}$$

$$(b) V_{NH} = V_{OH} - V_{IH}$$

$$6 = 10 - V_{IH}$$

$$V_{IH} = 4 \text{ V}$$

Since transistor in

saturation,  $V_o = 0.2 \text{ V} (V_{CE})$

$$V_B = 0.8 \text{ V} (V_{BE})$$

$$I_C = \frac{12 - 0.2}{2.2} = 5.3636 \text{ mA}$$

$$I_B = \frac{I_C}{\beta} = 0.17878 \text{ mA}$$

$$I_2 = \frac{0.8 + 12}{120} = 0.1067 \text{ mA}$$

$$I_1 = I_2 + I_B = 0.2854$$

$$\therefore \frac{V_{IH} - 0.8}{R} = 0.2854$$

$$R = 11.2 \text{ k}\Omega$$

$$(c) V_{in} = 12 \text{ V} \quad (R = 10 \text{ k})$$

since saturation,  $V_o = 0.2 \text{ V} (V_{OL})$

$$I_C = 5.3636 \text{ mA}$$

$$I_L = \frac{0.2 + 12}{100 + 120} = 0.0554$$

$$N = \left\lfloor \frac{I_C}{I_L} \right\rfloor = 96$$

$$V_{in} = 0.2 \text{ V}$$

since cutoff,  $V_o = 10 \text{ V} (V_{OH})$

$$I_C = \frac{12 - 10}{2.2} = 0.909 \text{ mA}$$

$$I_L = \frac{10 - 0.8}{100} = 0.092 \text{ mA}$$

$$N = \left\lfloor \frac{I_C}{I_L} \right\rfloor = 9$$

$$\min(96, 9) = 9 = \text{fanout}$$

### Solution Set B

$$(a) \frac{0.2 - (-12)}{10 + 80} = i = 0.135 \text{ mA} \quad (V_{in} = 0.2 \text{ V} \rightarrow \text{cutoff})$$

$$P = (0.2 + 12) i = 1.653 \text{ mW}$$

$$(b) V_{NH} = V_{OH} - V_{IH}$$

$$6 = 10 - V_{IH}$$

$$V_{IH} = 4 \text{ V}$$

since transistor in

saturation,  $V_o = V_{CE} = 0.2 \text{ V}$

$$V_B = 0.8 = V_{BE}$$

$$I_C = \frac{12 - 0.2}{2.2} = 5.3636 \text{ mA}$$

$$I_B = \frac{I_C}{\beta} = 0.17878 \text{ mA}$$

$$I_2 = \frac{0.8 + 12}{80} = 0.16 \text{ mA}$$

$$I_1 = I_2 + I_B = 0.33878 \text{ mA}$$

$$I_1 = \frac{V_{IH} - 0.8}{R}$$

$$R = 9.445 \text{ k}\Omega$$

$$(c) V_{in} = 12 \text{ V} \quad (R = 10 \text{ k})$$

$$I_C = 5.3636 \text{ mA}$$

$$I_L = \frac{0.2 - (-12)}{200 + 120}$$

$$= 0.05545 = 0.038125 \text{ mA}$$

$$N = \left\lfloor \frac{I_C}{I_L} \right\rfloor = 140$$

$$= \left\lfloor \frac{5.3636}{0.038125} \right\rfloor = 140$$

$$V_{in} = 0.2 \text{ V}$$

cutoff  
 $V_o = 10 = V_{OH}$

$$I_C = \frac{12 - 10}{2.2} = 0.909 \text{ mA}$$

$$I_L = \frac{10 - 0.8}{200} = 0.048 \text{ mA}$$

$$N = \left\lfloor \frac{I_C}{I_L} \right\rfloor = 19$$

$$\min(140, 19) = 19 = \text{fanout}$$

DTL:[JMD]

Question 3

[20]

For the DTL circuit in Figure 3 assume  $\beta_F = 65$ ,  $V_{BE(saturation)} = 0.8V$ , and  $V_{CE(saturation)} = 0.2V$ . The input voltages  $A$  and  $B$  can be  $0V$  or  $5V$

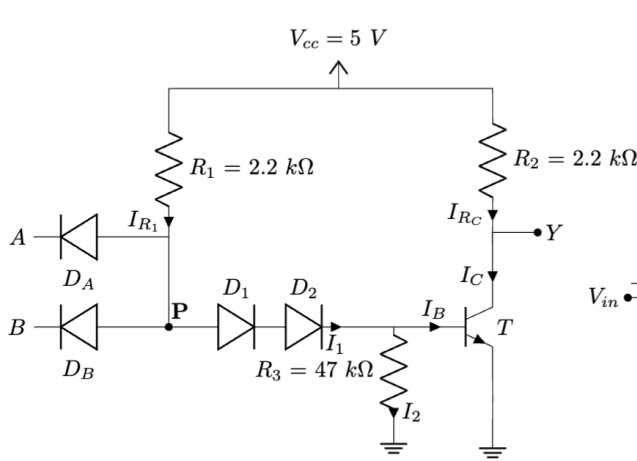


Figure 1: DTL NAND gate

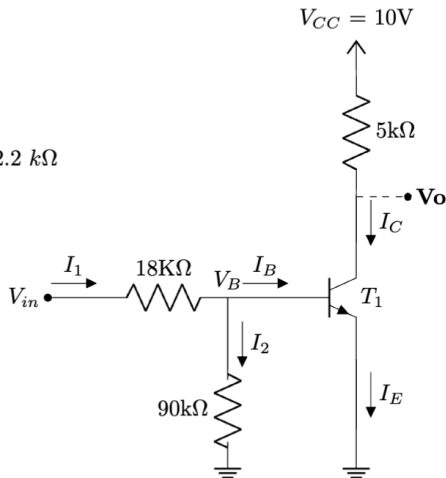


Figure 2: RTL for part (c) only

Only consider figure 1 for part (a) & part (b).

|     |     |                                                                                                                                                                               |     |
|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| CO2 | (a) | <b>Calculate</b> the maximum fanout when the DTL circuit is connected to identical loads, the same as the driver circuit in figure 1 and the inputs are set to $A = B = 5V$ . | [8] |
| CO2 | (b) | <b>Calculate</b> the total power dissipation of the DTL circuit in figure 1 when $A = 5$ & $B = 0$ .                                                                          | [5] |

The designer now wants to make an AND gate by using the DTL-NAND gate (figure: 1) as the driver and the RTL-NOT gate (figure 2) as the load.

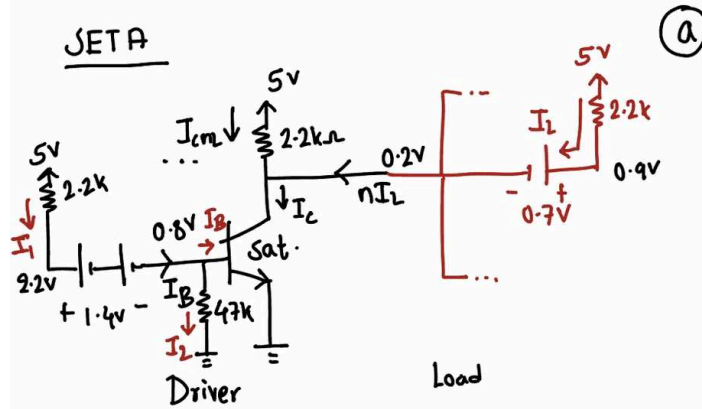
|     |     |                                                                                                                                                       |     |
|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| CO2 | (c) | <b>Calculate</b> the maximum number of RTL NOT gates a single DTL NAND gate can drive when the input of the DTL is both high ( $A = 5V$ , $B = 5V$ ). | [7] |
|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----|

Set A and Set B for this question are the same! 😊

Solution:

Question 03

JETA



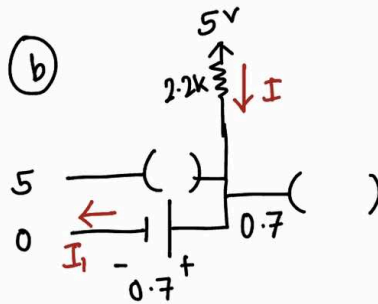
$$I_B = I_1 - I_2$$

$$I_B = \frac{5 - 0.2}{2.2k} - \frac{0.8}{47k} = 1.26 \text{ mA}$$

$$I_C = I_{C_{NL}} + n I_L$$

$$\beta I_B = \frac{5 - 0.2}{2.2k} + n \left( \frac{5 - 0.9}{2.2k} \right)$$

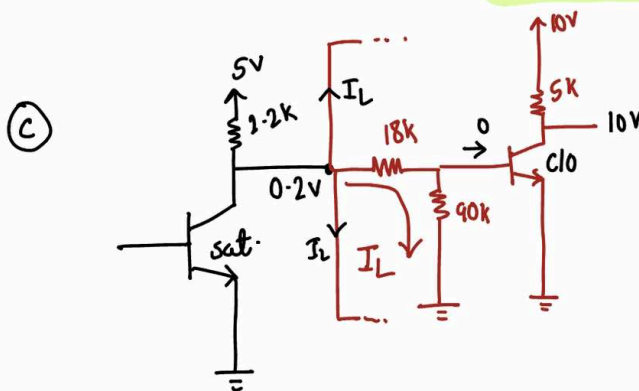
$$n = \lfloor 42.78 \rfloor = 42 \text{ (Ans)}$$



$$\text{Power} = (5 - 0) I$$

$$(P) = 5 \times \frac{5 - 0.7}{2.2k}$$

$$P = 9.77 \text{ mW (Ans)}$$



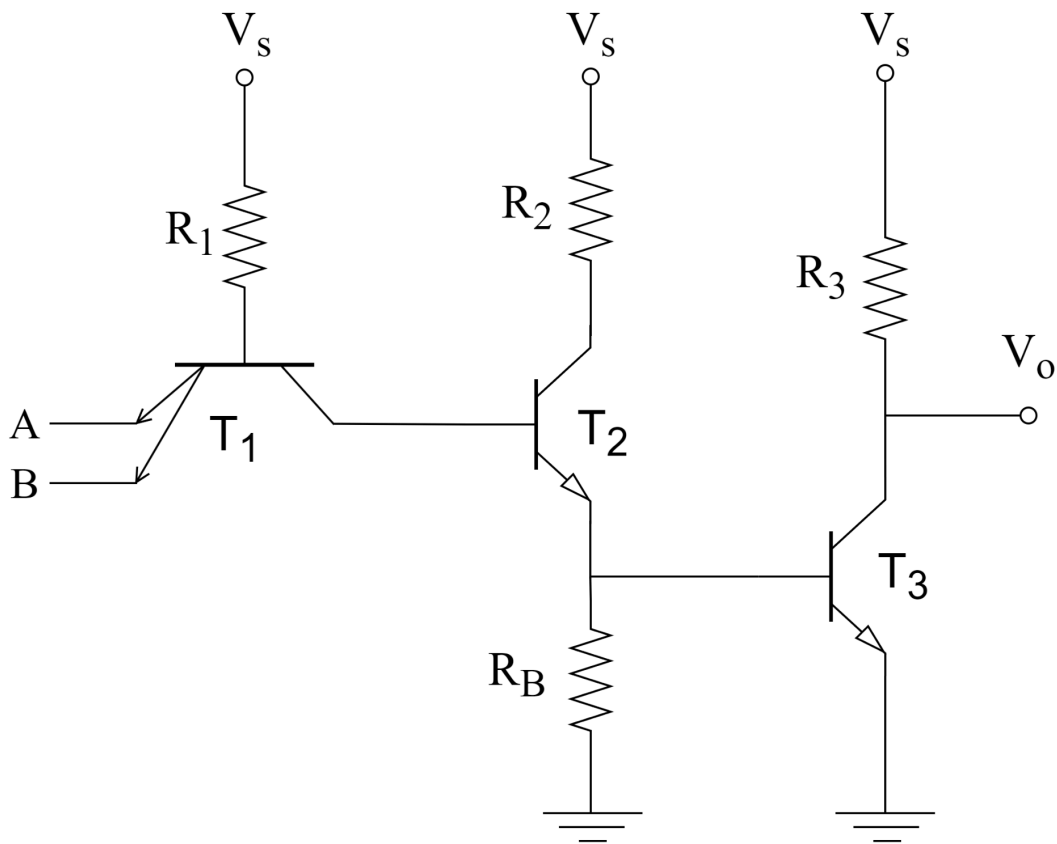
$$I_L = \frac{0.2 - 0}{(18 + 90)k} = 1.85 \times 10^{-3} \text{ mA}$$

$$n_{\max} = \frac{I_{C \max}}{I_L}$$

$$= \frac{5 - 0.2}{2.2k} = \frac{1179.4}{1.85 \times 10^{-3}}$$

$$n_{\max} \approx 1179$$

## TTL:[QSH]



**Set A:** [Solution: (a) 1V, (b) 1 & 5, (c) 5.1 V ]

Given,  $R_1 = 15.4\text{k}\Omega$ ,  $R_2 = 3\text{k}\Omega$ ,  $R_3 = 4.9\text{k}\Omega$ ,  $R_B = 0.5\text{k}\Omega$ ,  $V_S = 10\text{V}$

$V_V(\text{cut-in}) = 0.5\text{V}$ ,  $V_{CE}(\text{saturation}) = 0.2\text{V}$ ,  $V_{BE}(\text{saturation}) = 0.8\text{V}$ ,  $\beta_F = 30$ ,

$V_{BC}(\text{reverse active}) = 0.7\text{V}$ ,  $\beta_R = 0.1$

**Set B:** [Solution: (a) 1V, (b) 1 & 5, (c) 6.6 V ]

Given,  $R_1 = 21.4\text{k}\Omega$ ,  $R_2 = 4\text{k}\Omega$ ,  $R_3 = 6.4\text{k}\Omega$ ,  $R_B = 0.5\text{k}\Omega$ ,  $V_S = 13\text{V}$

$V_V(\text{cut-in}) = 0.5\text{V}$ ,  $V_{CE}(\text{saturation}) = 0.2\text{V}$ ,  $V_{BE}(\text{saturation}) = 0.8\text{V}$ ,  $\beta_F = 30$ ,

$V_{BC}(\text{reverse active}) = 0.7\text{V}$ ,  $\beta_R = 0.1$

|     |     |                                                                                                                                                                                                                                                                                   |    |
|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| (a) | CO1 | <b>Determine</b> the minimum amount of voltage required at the collector of $T_1$ to turn ON both transistors $T_2$ and $T_3$ simultaneously.                                                                                                                                     | 2  |
| (b) | CO1 | <b>Calculate</b> the minimum value of $\beta_F$ for transistors $T_2$ and $T_3$ such that this TTL logic gate can still function normally.                                                                                                                                        | 10 |
| (c) | CO1 | If 20 identical load gates similar to the given TTL circuit are connected to the output, <b>find</b> the output voltage when both inputs of the driver circuit are <b>Low</b> .<br>(Hint: One input of each load circuit is connected to $V_O$ , the other input is <b>High</b> ) | 8  |

Solution:

$$(a) \quad V_P = 0.5 + 0.5 = 1V$$

$$(b) \quad V_{B1} = 0.8 + 0.8 + 0.7 = 2.3V$$

$$I_{B1} = \frac{V_S - 2.3}{R_1} = 0.5 \text{ mA}$$

$$I_{E11} = I_{E12} = \beta_R I_{B1} = 0.05 \text{ mA}$$

$$I_{B2} = I_{C1} = I_{E11} + I_{E12} + I_{B1} = 0.6 \text{ mA}$$

$$I_{C2} = \frac{V_S - 1}{R_2} = 3 \text{ mA}$$

$$I_{E2} = I_{C2} + I_{B2} = 3.6 \text{ mA}$$

$$I' = \frac{0.8}{R_B} = 1.6 \text{ mA}$$

$$I_{B3} = I_{E2} - I' = 2 \text{ mA}$$

$$I_{C3} = \frac{V_S - 0.2}{R_3} = 2 \text{ mA}$$

$$\beta_{F2} = \frac{I_{C2}}{I_{B2}} = 5$$

$$\beta_{F3} = \frac{I_{C3}}{I_{B3}} = 1$$

$$(c) \quad \frac{V_S - V_O}{R_3} = 20 \times 0.05$$

$$\Rightarrow V_O = V_S - R_3$$

$$\text{For Set A: } V_O = 10 - 4.9 = 5.1V$$

$$\text{For Set B: } V_O = 13 - 6.4 = 6.6V$$

